লিমিট

- $\lim(x \pm y) = \lim(x) \pm \lim(y)$
- $\lim(xy) = \lim(x) \times \lim(y)$
- $lim\left(\frac{x}{y}\right) = \frac{\lim(x)}{\lim(y)}$
- $\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$
- $\lim(cx) = c.\lim(x)$
- $\lim(c) = c$
- $\lim_{\theta\to 0} \frac{\theta}{\sin\theta} = \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$
- $\lim_{\theta\to 0} \frac{\theta}{\sin k\theta} = \lim_{\theta\to 0} \frac{\sin k\theta}{\theta} = k$
- $\lim_{\theta\to 0} \frac{\theta}{\tan\theta} = \lim_{\theta\to 0} \frac{\tan\theta}{\theta} = 1$
- $\lim_{\theta\to 0} \frac{\theta}{\sin^{-1}\theta} = \lim_{\theta\to 0} \frac{\sin^{-1}\theta}{\theta} = 1$
- $\lim_{\theta\to 0} \frac{\theta}{\tan^{-1}\theta} = \lim_{\theta\to 0} \frac{\tan^{-1}\theta}{\theta} = 1$
- $\lim_{\theta\to 0} \frac{k\theta}{\tan k\theta} = \lim_{\theta\to 0} \frac{\tan k\theta}{\theta} = k$
- $\lim_{\theta\to 0} \frac{\sin^2\theta}{\theta^2} = 1$
- $\lim_{\theta\to 0} \frac{\sin^2\theta}{\theta} = \theta$
- $\lim_{\theta\to 0} \frac{\sin^n k\theta}{\theta^n} = k^n$
- $\lim_{x\to 0} \frac{e^x-1}{x} = 1$
- $\lim_{x\to 0} (1+mx)^{\frac{n}{x}} = e^{mn}$
- $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to 0} \frac{(1+x)^n-1}{x} = n$
- $\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$
- $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^x = e$
- $\lim_{x\to\infty} \left(1+\frac{m}{x}\right)^x = e^m$
- $\lim_{x\to\infty} \left(1+\frac{m}{x}\right)^{nx} = e^{mn}$
- $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \frac{d}{dx}f(x) = \frac{dy}{dx} = f'(x)$
- $\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$
- $\lim_{x\to a} \frac{a^x-1}{x} = \ln a$

অন্তরীকরণ

$\frac{d}{dx}(c) = 0$

$\frac{d}{dx}(x) = 1$

$\frac{d}{dx}(x^n) = n.x^{n-1}$

$\frac{d}{dx}(u \pm v) = \frac{d}{dx}(u) \pm \frac{d}{dx}(v)$

$\frac{d}{dx}(uv) = u\frac{d}{dx}(v) + v\frac{d}{dx}(u)$

$\frac{d}{dx}(\ln x) = \frac{1}{x}\ ;\ (x > 0)$

$\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(\tan x) = \frac{1}{\cos^2 x} = \sec^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$

$\frac{d}{dx}(\cot x) = -\frac{1}{\sin^2 x} = -\text{cosec}^2 x$

$\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}(\cos^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}(uvw) = uvw\left\{\frac{\frac{d}{dx}(u)}{u} + \frac{\frac{d}{dx}(v)}{v} + \frac{\frac{d}{dx}(w)}{w}\right\}$

$\frac{d}{dx}(\log_a x) = \frac{1}{x\ln a}; (x > 0)$

$\frac{d}{dx}(a^x) = a^x \ln a$

$\frac{d}{dx}(e^x) = e^x$

$\frac{d}{dx}(e^{mx}) = me^x$

$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$

$\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

$\frac{d}{dx}(\cot^{-1}x) = -\frac{1}{1+x^2}$

$\frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$

$\frac{d}{dx}(\text{cosec}^{-1}x) = -\frac{1}{x\sqrt{x^2-1}}$

$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{d}{dx}(u) - u\frac{d}{dx}(v)}{v^2}$

$\frac{dy}{dx} = \frac{dy}{dz}.\frac{dz}{dx}$

f(x) এর কোন পদের আকার	প্রতিস্থাপন
$\sqrt{x^2-a^2}$	$x = a.sec\theta$ অথবা $x = a.cosec\theta$
$\sqrt{a^2-x^2}$	$x = a.sin\theta$ অথবা $x = a.cos\theta$
$\sqrt{a^2+x^2}$	$x = a.tan\theta$ অথবা $x = a.cot\theta$
$\sqrt{a^2+x^2}$ এবং $\sqrt{a^2-x^2}$	$x^2 = a^2\cos 2\theta$
$\sqrt{a+x}$ অথবা $\sqrt{a-x}$	$x = a\cos 2\theta$
$\frac{2x}{1-x^2}$ অথবা $\frac{2x}{1-x^2}$ অথবা $\frac{1-x^2}{1+x^2}$	$x = tan\theta$ অথবা $x = cot\theta$
$\frac{1+x}{1-x}$ অথবা $\frac{1-x}{1+x}$	$x = tan\theta$
$\sqrt{\frac{1+x}{1-x}}$ অথবা $\sqrt{\frac{1-x}{1+x}}$	$x = cos\theta$ অথবা $x = sin\theta$
$\sqrt{\frac{x-1}{x+1}}$ অথবা $\sqrt{\frac{x+1}{x-1}}$	$x = sec\theta$ অথবা $x = cosec\theta$
$\sqrt{\frac{1+x}{2}}$ অথবা $\sqrt{\frac{1-x}{2}}$	$x = cos\theta$
$\sqrt{x}+\sqrt{1-x}$	$x = \sin^2\theta$ অথবা $x = \cos^2\theta$
$(1+x^2)^n$	$x = tan\theta$ অথবা $x = cot\theta$
$(1-x^2)^n$	$x = sin\theta$ অথবা $x = cos\theta$
$(x^2-1)^n$	$x = sec\theta$ অথবা $x = cosec\theta$

<u>n তম অন্তরজঃ</u>

1. $y = x^n$হলে, $y_n = n!$
2. $y = e^{ax+b}$ হলে, y
3. $y = a^x$ হলে, $y_n = a^x(\ln a)^n$
4. $y = \sin(ax+b)$ হলে, $y_n = a^n\sin\left[\frac{n\pi}{2} + ax + b\right]$
5. $y = \cos(ax+b)$ হলে, $y_n = y_n = a^n\cos\left[\frac{n\pi}{2} + ax + b\right]$
6. $y = \ln x$ হলে, $y_n = \frac{(-1)^{n-1}(n-1)!}{x^n}$
7. $y = a^{ax}T(bc+x)$ হলে,
   $y_n = (a^2+b^2)^{\frac{n}{2}}.e^{ax}T(bx+c+n\tan^{-1}\frac{b}{a})$;[যেখানে T=sinঅথবা cos]

যোগজীকরন

- $\int x^n dx = \frac{x^{n+1}}{n+1} + c; (n \neq -1)$
- $\int dx = x + c$
- $\int \frac{1}{x}dx = \ln x + c$
- $\int \frac{1}{\sqrt{x}}dx = 2\sqrt{x}$
- $\int e^x dx = e^x + c$
- $\int e^{mx}dx = \frac{1}{m}e^{mx} + c$
- $\int a^x dx = \frac{a^x}{\ln a} + c$
- $\int \ln x dx = x\ln x - x + c$
- $\int \sin x\ dx = -\cos x + c$
- $\int \sin mx\ dx = -\frac{1}{m}\cos mx + c$
- $\int \cos x\ dx = \sin x + c$
- $\int \tan x\ dx = -\ln(\cos x) + c = \ln(\sec x) + c$
- $\int \cot x\ dx = \ln(\sin x) + c = -\ln(\text{cosec}\, x) + c$
- $\int \sec x\ dx = \ln(\sec x + \tan x) + c = \ln\left\{\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right\} + c = \ln\left|\frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}}\right| + c$
- $\int \text{cosec}\, x\, dx = \ln(\text{cosec}\, x - \cot x) + c = -\ln(\text{cosec}\, x + \cot x) + c$
- $\int \text{cosec}\, x\, dx = \ln\left(\tan\frac{x}{2}\right) + c$
- $\int \sec^2 x\ dx = \tan x + c$
- $\int \text{cosec}^2 x\ dx = -\cot x + c$
- $\int \sec x.\tan x\ dx = -\sec x + c$
- $\int \text{cosec}\, x.\cot x\ dx = -\text{cosec}\, x + c$

- $\int \frac{1}{\sqrt{1-x^2}}dx = \sin^1 x + c$
- $\int \frac{1}{\sqrt{1-x^2}}dx = -\cos^1 x + c$
- $\int \frac{dx}{1+x^2} = \tan^{-1} x + c$
- $\int \frac{dx}{1+x^2} = \cot^{-1} x + c$
- $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + c$
- $\int \frac{dx}{x\sqrt{x^2-1}} = -\mathrm{cosec}^{-1} x + c$
- $\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$; [x = a. tan θ]
- $\int \frac{1}{x^2-a^2}dx = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + c$
- $\int \frac{1}{a^2-x^2}dx = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + c$
- $\int \frac{1}{\sqrt{a^2-x^2}}dx = \sin^1\frac{x}{a} + c$; [x = a. sin θ]
- $\int \frac{1}{\sqrt{x^2+a^2}}dx = \ln\left(x+\sqrt{x^2+a^2}\right) + c$; [ধরি, $x = a\tan\theta$]
- $\int \frac{1}{\sqrt{x^2-a^2}}dx = \ln\left(x+\sqrt{x^2-a^2}\right) + c$; [ধরি, $x = a\sec\theta$]
- $\int \frac{dx}{x\sqrt{x^2+a^2}} = \frac{1}{2a}\ln\sqrt{x^2+a^2+a} + c = \frac{1}{a}\ln\left(\frac{x}{\sqrt{x^2+a^2}+a}\right) + c$
- $\int \sqrt{a^2-x^2}dx = \frac{1}{2}x\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$

- $\int \sqrt{\frac{a+x}{a-x}}dx = a\sin^{-1}\frac{x}{a} - \sqrt{a^2-x^2} + c$
- $\int \sqrt{\frac{a-x}{a+x}}dx = a\sin^{-1}\frac{x}{a} + \sqrt{a^2-x^2} + c$
- $\int \frac{xdx}{(x^2+a^2)^{3/2}} = -\frac{1}{\sqrt{x^2+a^2}} + c$
- $\int \frac{dx}{(x^2+a^2)^{3/2}} = \frac{-x}{a^2\sqrt{x^2+a^2}} + c$ [To proof $x = a\tan\theta$]
- $\frac{dx}{(x^2+a^2)^2} = \frac{1}{2a^3}\left[\tan^{-1}\frac{x}{a} + \frac{ax}{a^2+x^2}\right] + c$
- $\int e^{mx}\{mf(x) + f'(x)\}dx = e^{mx}f(x) + c$
- $\int f(x)+xf'(x)dx = xf(x) + c$
- $\int \{f(x)\}^n.f'(x)dx = \frac{\{f(x)\}^{n+1}}{n+1} + c$
- $\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$
- $\int \frac{f'(x)}{\sqrt{f(x)}}dx = 2\sqrt{f(x)} + c$
- $\int \frac{f(x)}{1+\{f(x)\}^2} = \tan^{-1} f(x) + c$
- $\int \frac{f'(x)}{a^2+\{f(x)\}^2} = \frac{1}{a}\tan^{-1}\left(\frac{f(x)}{a}\right) + c$
- $\int \frac{f'(x)}{a^2-\{f(x)\}^2} = \frac{1}{2a}\ln\left|\frac{a-f(x)}{a+f(x)}\right| + c$
- $\int \frac{f'(x)}{\sqrt{a^2-\{f(x)\}^2}} = \sin^{-1}\left(\frac{f(x)}{a}\right) + c$
- $\int \frac{f'(x)}{f(x)\sqrt{\{f(x)\}^2-a^2}}dx = \frac{1}{a}sec^{-1}\left(\frac{f(x)}{a}\right) + c$
- $\int \frac{f'(x)dx}{f(x)\sqrt{\{f(x)\}^2-1}} = \sec^{-1}f(x) + c$
- $\int e^{f(x)}f'(x)dx = e^{f(x)} + c$

- $\int \{f(x) + xf'(x)\}dx = xf(x) + c$
- $\int \cos\{f(x)\}f'(x)dx = \sin\{f(x)\} + c$
- $\int \sin\{f(x)\}f'(x)dx = -\cos\{f(x)\} + c$
- $\int \sec^2\{f(x)\}f'(x)dx = \tan\{f(x)\} + c$
- $\int \sec\{f(x)\}\tan\{f(x)\}f'(x)dx = \sec\{f(x)\} + c$
- $\int \sqrt{x^2+a^2}dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left|x+\sqrt{x^2+a^2}\right| + c$; [ধরি, x = a. tan θ]
- $\int \sqrt{x^2-a^2}dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + c$; [ধরি, x = a. tan θ]
- $\int \sqrt{x^2-a^2}dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$
- $\int \sqrt{a^2-x^2}dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$
- $\int uv\,dx = u\int v\,dx - \int\left\{\frac{d}{dx}(u)\int vdx\right\}dx$
- $\int e^{ax}\{a.f(x) + f'(x)\}\,dx = e^{ax}.f(x) + c$
- $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + c$
- $\int e^{ax}\cos bx\,dx = \int \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\cos bx) + c$

- $\sin(A+B) = \sin A\cos B + \cos A\sin B$
- $\sin(A-B) = \sin A\cos B - \cos A\sin B$
- $\cos(A+B) = \cos A\cos B - \sin A\sin B$
- $\cos(A-B) = \cos A\cos B + \sin A\sin B$
- $\tan(A+B) = \frac{\tan A+\tan B}{1-\tan A,\tan B}$
- $\tan(A-B) = \frac{\tan A-\tan B}{1+\tan A,\tan B}$
- $\tan(A+B+C) = \frac{tanA+tanB+tanC-tanA.tanB.tanC}{1-tanA.tanB-tanB.tanC-tanC.tanA}$
- $\cot(A+B) = \frac{\cot A\cot B-1}{\cot A+\cot B}$
- $\cot(A-B) = \frac{\cot A\cot B+1}{\cot A-\cot B}$
- $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{(A-B)}{2}$
- $\sin A - \sin B = 2\cos\frac{(A+B)}{2}\sin\frac{(A-B)}{2}$
- $\cos A + \cos B = 2\cos\frac{(A+B)}{2}\cos\frac{(A-B)}{2}$
- $\cos A - \cos B = 2\sin\frac{(A+B)}{2}\sin\frac{(B-A)}{2}$
- $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$
- $\sin(A+B)\sin(A-B) = \cos^2 B - cos^2 A$
- $\cos(A+B)\cos(A-B) = cos^2 A - \sin^2 B$
- $\cos(A+B)\cos(A-B) = cos^2 B - \sin^2 A$
- $\sin(A+B) + \sin(A-B) = 2\sin A\cos B$
- $\cos(A-B) + \cos(A+B) = 2\cos A\cos B$

- $\sin(A+B)-\sin(A-B)=2\cos A\sin B$
- $\cos(A-B)-\cos(A+B)=2\sin A\sin B$
- $sinC+sinD=2\sin\frac{(C+D)}{2}cos\frac{(C-D)}{2}$
- $sinC-sinD=2\cos\frac{(C+D)}{2}sin\frac{(C-D)}{2}$
- $cosC+cosD=2\cos\frac{(C+D)}{2}cos\frac{(C-D)}{2}$
- $cosC-cosD=2\sin\frac{(C+D)}{2}sin\frac{(D-C)}{2}$
- $\sin 2A=2\sin A\cos A$
- $\sin 2A=\frac{2\tan A}{1+\tan^2 A}$
- $\cos 2A=\cos^2 A-\sin^2 A$
- $\cos 2A=2\cos^2 A-1$
- $\cos 2A=1-2\sin^2 A$
- $\cos 2A=\frac{1-\tan^2 A}{1+\tan^2 A}$
- $2\sin^2 A=1-cos2A$
- $2\cos^2 A=1+cos2A$
- $\tan 2A=\frac{2\tan A}{1-\tan^2 A}$
- $\tan 3A=\frac{3\tan A-\tan^3 A}{1-3\tan^2 A}$
- $4\sin^3 A=3\sin A-\sin 3A$
- $4\cos^3 A=3\cos A+\cos 3A$
- $\tan^3 A=3\tan A+(3\tan^2 A-1)\tan 3A$

- $\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$
- $\cot^{-1}x+\tan^{-1}x=\frac{\pi}{2}$
- $\cos^{-1}x+\text{cosec}^{-1}x=\frac{\pi}{2}$
- $\sin^{-1}x+\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}+y\sqrt{1-x^2}\right]$
- $\sin^{-1}x-\sin^{-1}y=\sin^{-1}\left[x\sqrt{1-y^2}-y\sqrt{1-x^2}\right]$
- $\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\sqrt{1-y^2}\right]$
- $\cos^{-1}x-\cos^{-1}y=\cos^{-1}\left[xy+\sqrt{1-x^2}\sqrt{1-y^2}\right]$
- $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$
- $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\frac{x-y}{1+xy}$
- $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\frac{x+y+z-xyz}{1-xy-yz-zx}$
- $\sin^{-1}\frac{1}{x}=cosec^{-1}x$
- $\cos^{-1}\frac{1}{x}=\cot^{-1}x$
- $\tan^{-1}\frac{1}{x}=\cot^{-1}x$
- $\sin^{-1}(-x)=-\sin^{-1}x$
- $\cos^{-1}(-x)=\pi-\cos^{-1}x$
- $tan^{-1}(-x)=-\tan^{-1}x$
- $\cot^{-1}(-x)=-\cot^{-1}x$
- $\sec^{-1}(-x)=\pi-sec^{-1}x$
- $cosec^{-1}(-x)=-\text{cosec}^{-1}x$
- $2\tan^{-1}x=\tan^{-1}\frac{2x}{1-x^2}=\cos^{-1}\frac{1-x^2}{1+x^2}=\sin^{-1}\frac{2x}{1+x^2}$
- $2\sin^{-1}x=\sin^{-1}\left(2x\sqrt{1-x^2}\right)$
- $2\cos^{-1}x=\cos^{-1}(2x^2-1)$
- $3\sin^{-1}x=\sin^{-1}(3x-4x^3)$
- $3\cos^{-1}x=\cos^{-1}(4x^3-3x)$
- $3\tan^{-1}x=\tan^{-1}\frac{3x-x^3}{1-3x^2}$

- $\frac{1}{2}\sin^{-1}x=\tan^{-1}\frac{1-\sqrt{1-x^2}}{x}$
- $\frac{1}{2}\cos^{-1}x=\cos^{-1}\sqrt{\frac{1+x}{2}}=\sin^{-1}\sqrt{\frac{1-x}{2}}=\tan^{-1}\sqrt{\frac{1-x}{1+x}}$
- $\frac{1}{2}\tan^{-1}x=\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$
- $\tan^{-1}\frac{ax+by}{ax-by}=\frac{\pi}{4}+\tan^{-1}\frac{by}{ax}$
- $\tan^{-1}\frac{ax+by}{bx-ay}=\tan^{-1}\frac{a}{b}+\tan^{-1}\frac{y}{x}$
- $\tan^{-1}\frac{ax-by}{bx+ay}=\tan^{-1}\frac{a}{b}-\tan^{-1}\frac{y}{x}$
- $\tan^{-1}\frac{1+x}{1-x}=\frac{\pi}{4}+\tan^{-1}x$
- $\tan^{-1}\frac{1-x}{1+x}=\frac{\pi}{4}-\tan^{-1}x$
- $\sin^{-1}x=\cos^{-1}\sqrt{1-x^2}=\tan^{-1}\frac{x}{\sqrt{1-x^2}}=\cot^{-1}\frac{\sqrt{1-x^2}}{x}=\sec^{-1}\frac{1}{\sqrt{1-x^2}}=\text{cosec}^{-1}\frac{1}{x}$
- $\sin\theta=\sin\alpha$ হলে $\theta=n\pi+(-1)^n\alpha$
- $\cos\theta=\cos\alpha$ হলে $\theta=2n\pi\pm\alpha$
- $\tan\theta=\tan\alpha$ হলে $\theta=n\pi+\alpha$
- $\sin\theta=0$ হলে $\theta=n\pi+a$
- $\cos\theta=0$ হলে $\theta=(2n+1)\frac{\pi}{2}$
- $\tan\theta=0$ হলে $\theta=n\pi$
- $\sin\theta=1$ হলে $\theta=(4n+1)\frac{\pi}{2}$
- $\sin\theta=-1$ হলে $\theta=(4n-1)\frac{\pi}{2}$
- $\cos\theta=1$ হলে $\theta=2n\pi$
- $\cos\theta=-1$ হলে $\theta=(2n+1)\pi$

সরলরেখা

- কোন বিন্দুর কার্তেসীয় স্থানাংক (x,y) এবং পোলার স্থানাংক (r,θ) হলে,

  $y=r\sin\theta$

  $x=r\cos\theta$

  $r=\sqrt{x^2+y^2}$

  $\theta=\tan^{-1}\frac{y}{x}$

- $A(x_1,y_1),B(x_2,y_2)$ হলে AB= $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$
- $A(x,y)$ বিন্দু থেকে, x অক্ষের দূরত্ব = |x|

  y অক্ষের দূরত্ব=|y|

- $A(x_1,y_1),B(x_2,y_2)$ এর মধ্যবিন্দুর স্থানাঙ্ক $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2},\right)$
- $A(x_1,y_1),B(x_2,y_2),C(x_3,y_3)$ ত্রিভুজের তিনটি শীর্ষ হলে,

  ক্ষেত্রফল= $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}=\frac{1}{2}[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)]$

  ভরকেন্দ্র $\equiv\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$

- কোন ত্রিভুজের তিনটি বাহু BC,CA,AB এর মধ্যবিন্দু যথাক্রমে $D(x_1,y_1),E(x_2,y_2),F(x_3,y_3)$ হলে,

  $A\equiv(x_3+x_2-x_1,y_3+y_2-y_1)$

  $B\equiv(x_1+x_3-x_2,y_1+y_3-y_2)$

  $C\equiv(x_1+x_2-x_3,y_1+y_2-y_3)$

  কোন সামান্তরিকের তিনটি শীর্ষ $A(x_1,y_1),B(x_2,y_2),C(x_3,y_3)$ হলে,

  ৪র্থ শীর্ষ,$D\equiv(x_1+x_3-x_2,y_1+y_3-y_2)$

- $a_1x+b_1y+c_1=0,a_2x+b_2y+c_2=0,a_3x+b_3y+c_3=0$ রেখা তিনটি সমবিন্দু হবে

  যদি, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}=0$

- $a_1x+b_1y+c_1=0$ ও $a_2x+b_2y+c_2=0$ রেখা ২টি,

  একই রেখা হবে যদি, $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

  সমান্তরাল রেখা হবে যদি, $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$

  লম্ব হবে যদি, $\frac{a_1}{a_2}=-\frac{b_2}{b_1}$ হয়।